2.4.2.2 Predictor variables for the shallow landslide susceptibility assessment

Based on the work of Depicker et al. (2020a), we selected the following predictor variables for the shallow landslide susceptibility assessment (Table 1):

- Slope;
- Planar curvature;
- Profile curvature;
- North exposedness;
- East exposedness;
- Peak ground acceleration (PGA, a measure of seismic activity);
- Distance to faults;
- Distance to rivers;
- Land use/cover;
- Presence in the rejuvenated landscape (i.e. the influence of the formation of the Rift mountains on the river erosion of the landscape – see Text Box 2 and Figure 3); and Lithology.
- Additionally, we used the elevation, the presence of roads within 50 m, and the clay content of the soil. We distinguished six (6) land use/cover types: closed agriculture, open agriculture, irrigated lands, forests, open lands, and built-up land. We distinguished five (5) types of lithology: schists, granites, quartzites, volcanic rocks, and alluvial depositions.

These natural and human-related environmental predictors are environmental characteristics that could explain the occurrence of the landslides. The environmental characteristics were extracted for the landslide source area. Statistical criteria assessed to validate the use the predictor variables listed in table below and are presented in Appendix 1.

Table 1: **Landslide predictor variables evaluated for the susceptibility analysis, themes according to Reichenbach et al. (2018)**

Theme	Predictor variable	Units	Resolution/scale	Source
Morphology	Elevation	m	10, 30	DEM Rwanda, SRTM
	Slope	°	10, 30	
	Profile curvature	m^{-1}	10, 30	
	Planar curvature	m^{-1}	10, 30	
	North exposure	–	10, 30	
	East exposure	–	10, 30	
	Presence in rejuvenated landscape	–	1/1000	Depicker et al. (2020a)
Hydrology	Distance to rivers	m	10, 30	DEM Rwanda, SRTM
Geology	Lithology	–		
	Distance to faults	m	10, 30	Smets et al. (2016) and Delvaux et al. (2017)
	Peak ground acceleration (PGA)	$m\ s^{-2}$	2.2 km	Delvaux et al. (2017)
Land cover	Land use/cover types	–		GGGI / RWB
	Distance to roads within 50 m	m		RTDA
	Clay content (soil with >35% clay)	% clay		